

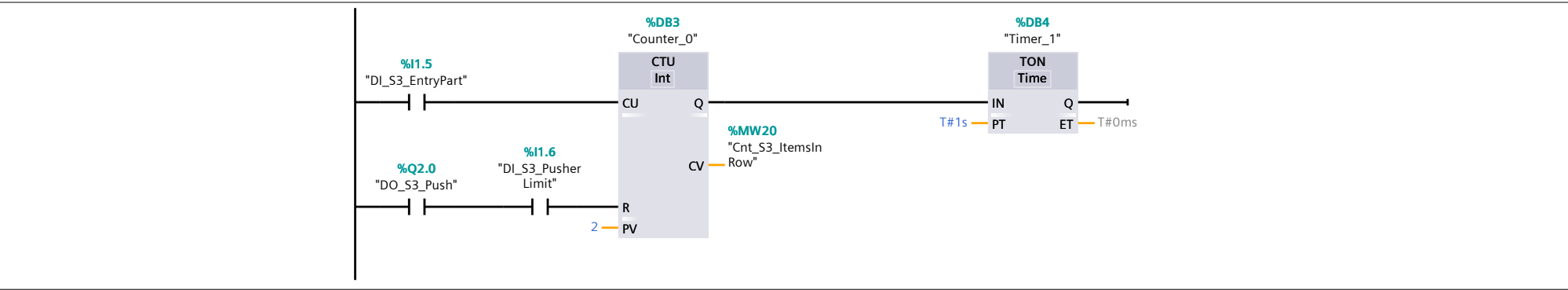
Station3 [FC3]

Station3 Properties							
General							
Name	Station3	Number	3	Type	FC	Language	LAD
Numbering	Automatic						
Information							
Title	Station 3 — Packaging & Palletizing	Author	Youness Tebbaq	Comment	Station 3 (Packaging & Pal- letizing). -Final products are pushed into an elevator two at a time (Cnt_S3_ItemsInRow). -After a 1-second debounce the pusher activates. The pallet is loaded with a row of items, the elevator raises to its upper limit, a clamp/plate sequence is executed (two iterations), and the elevator lowers again. -When a finished pallet reaches the exit sensor, the outfeed roller and remover start. The sequence uses three counters (counter_0 for items, counter_1 for the clamp/plate cycle, counter_2 for the elevator lower step) and three edge-trigger flags (b_S3_Trig_PalletLoaded, b_S3_Trig_OpenPlate, b_S3_Trig_PlateCycle).	Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
▼ Temp			
Tage_10	Bool		
rs_db	Bool		
Constant			
▼ Return			
Station3	Void		

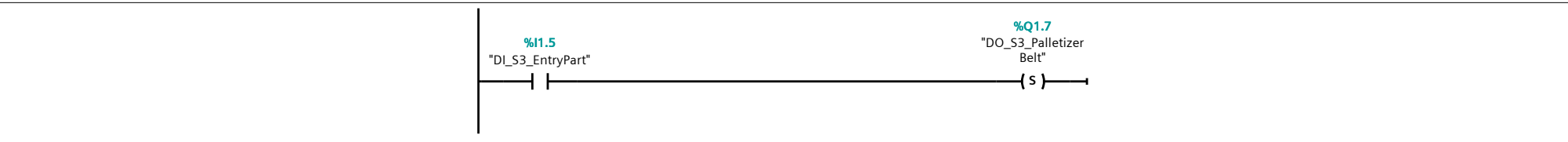
Network 1: Count items entering the row

Counter_0 (Cnt_S3_ItemsInRow, CTU instance) counts the parts entering the station on the entry diffuse sensor (DI_S3_EntryPart). When the count reaches 2 (PV=2), the counter's Q output goes high and triggers the debounce timer in Network 2. The pusher limit sensor (DI_S3_PusherLimit) resets the counter when the pusher retracts, so the next row can be counted.



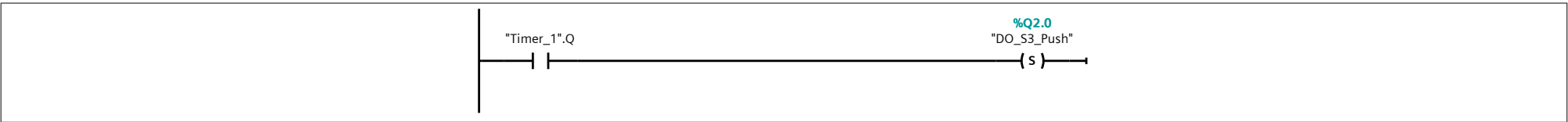
Network 2: 1-second debounce after the row is counted

TON timer (Inst_S3_TonRowDebounce, PT=1s) starts when the counter reaches 2. After 1 second the timer's Q output goes high and triggers the pusher in Network 3. This debounce prevents the pusher from firing on a false count during fast part arrival.



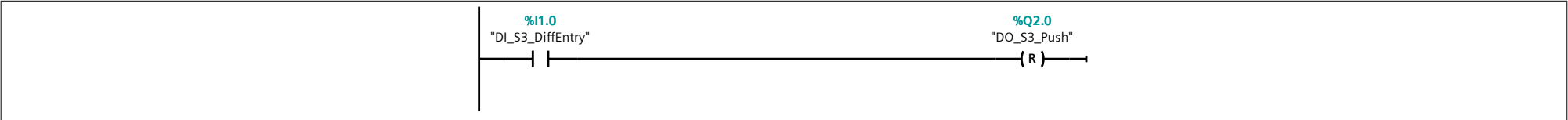
Network 3: Extend the pusher after the debounce

When the debounce timer elapses, energise the pusher coil (SET DO_S3_Push). The pusher extends and pushes the row of items into the elevator. The pusher is reset in Network 4 by the entry diffuse sensor.



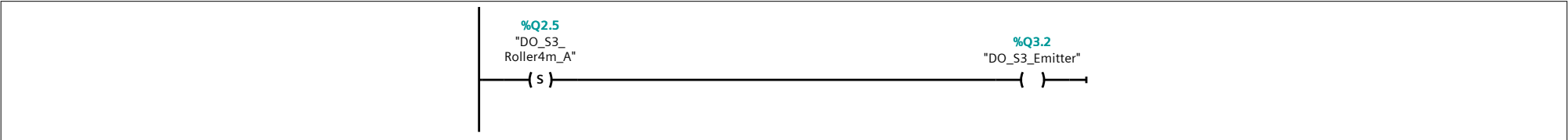
Network 4: Retract the pusher when the item passes the entry sensor

When the items have been pushed past the diffuse sensor (DI_S3_DiffEntry), reset the pusher coil. The pusher retracts and the counter is ready to count the next row.



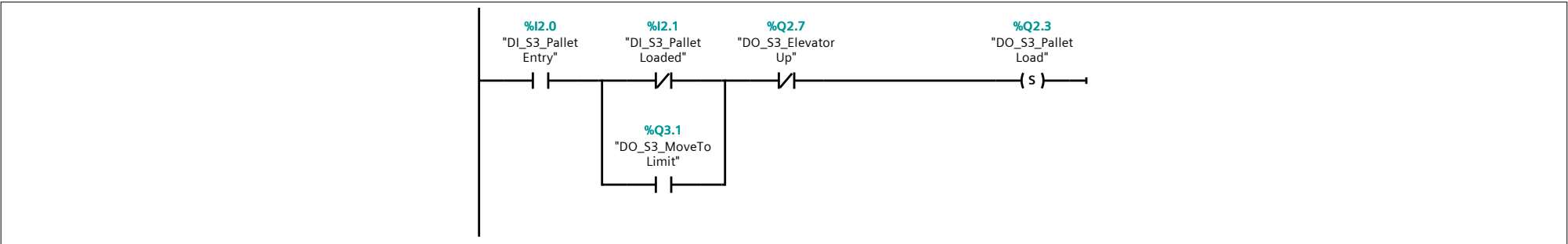
Network 5: Always run — Emitter 2 and roller A

Emitter 2 (DO_S3_Emitter) and roller conveyor A (DO_S3_Roller4m_A) are permanently ON. Emitter 2 spawns pallets to hold final products onto the roller; roller A continuously carries them toward the elevator. Roller B is started only when a pallet exits (Network 18).



Network 6: Start the pallet load sequence

When a pallet enters (DI_S3_PalletEntry) AND it is not already loaded (NOT DI_S3_PalletLoaded) AND the elevator is not raised (NOT DO_S3_ElevatorUp) OR the arm is not set to move to its limit yet (DO_S3_MoveToLimit), set the pallet-load flag (SET DO_S3_PalletLoad).

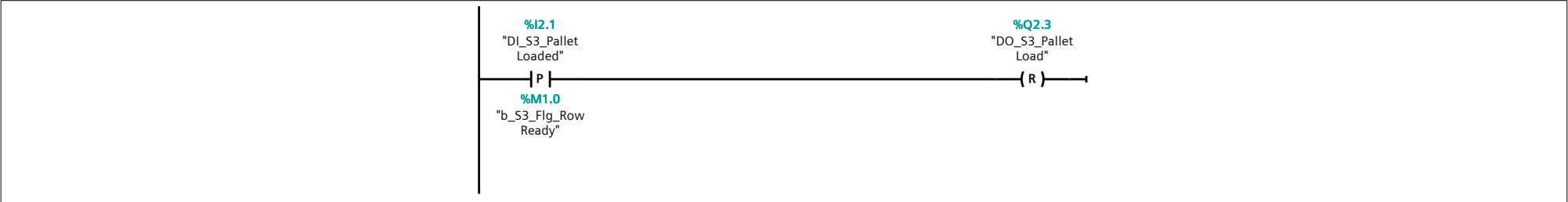


Network 7: Clear the pallet load when the row is ready

Clear the pallet-load flag (RESET DO_S3_PalletLoad) when:

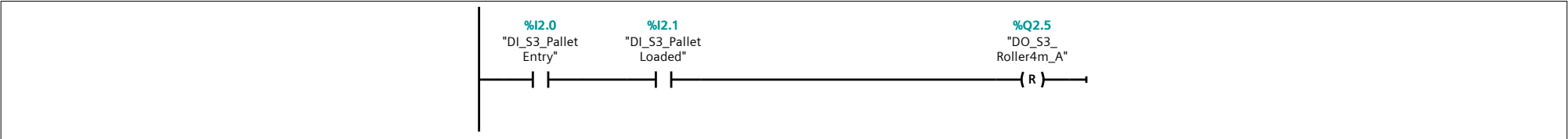
- the pallet is fully loaded (DI_S3_PalletLoaded), AND
- the row is ready flag is set (b_S3_Flg_RowReady).

The row-ready flag must be set elsewhere in the program (or via HMI) to confirm the row has been properly placed on the pallet.



Network 8: Stop roller A when the pallet is loaded

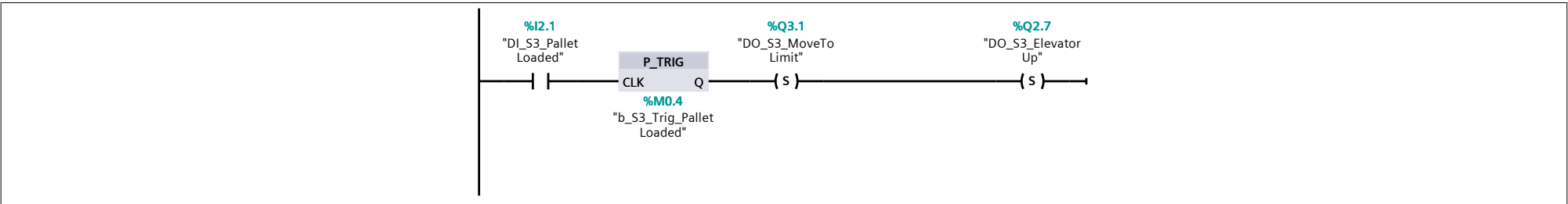
When a pallet enters (DI_S3_PalletEntry) AND it has been loaded (DI_S3_PalletLoaded), stop roller A (RESET DO_S3_Roller4m_A). This prevents new pallets from being pushed onto the elevator before that is ready.



Network 9: Start the elevator-up sequence on rising edge of pallet-loaded

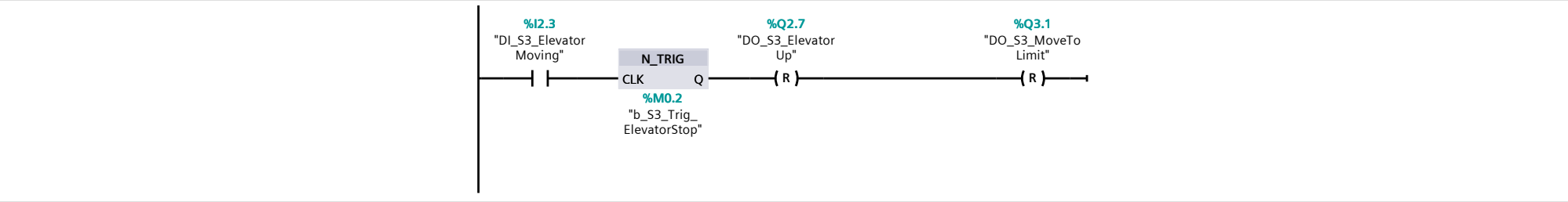
On the rising edge of the pallet-loaded sensor (P_TRIG b_S3_Trig_PalletLoaded), energise BOTH the elevator-up coil (SET DO_S3_ElevatorUp) AND the move-to-limit coil (SET DO_S3_MoveToLimit).

The elevator starts raising and the arm drives to its upper position.



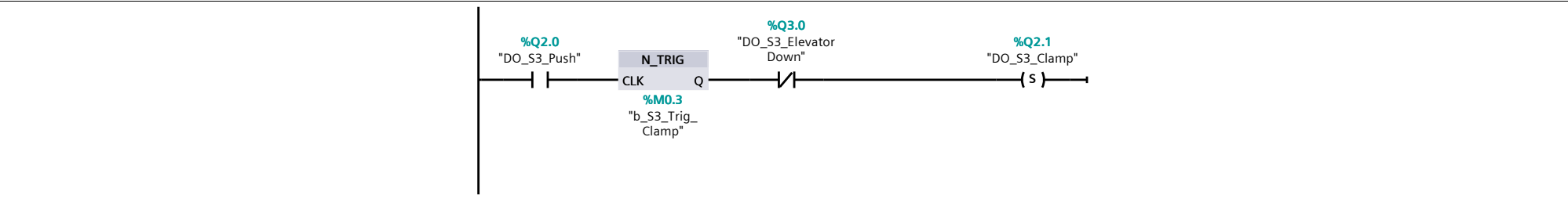
Network 10: Stop the elevator when motion ends

On the falling edge of the elevator-moving sensor (N_TRIG b_S3_Trig_ElevatorStop), reset BOTH the elevator-up coil AND the move-to-limit coil. The elevator stops at its upper position flags.



Network 11: Start the clamp sequence on falling edge of push

On the falling edge of the push coil (N_TRIG b_S3_Trig_Clamp), AND the elevator isn't down coil (NOT DO_S3_ElevatorDown) set the clamp coil (SET DO_S3_Clamp). The clamp closes to grip the row of items while the elevator remains at its upper position.



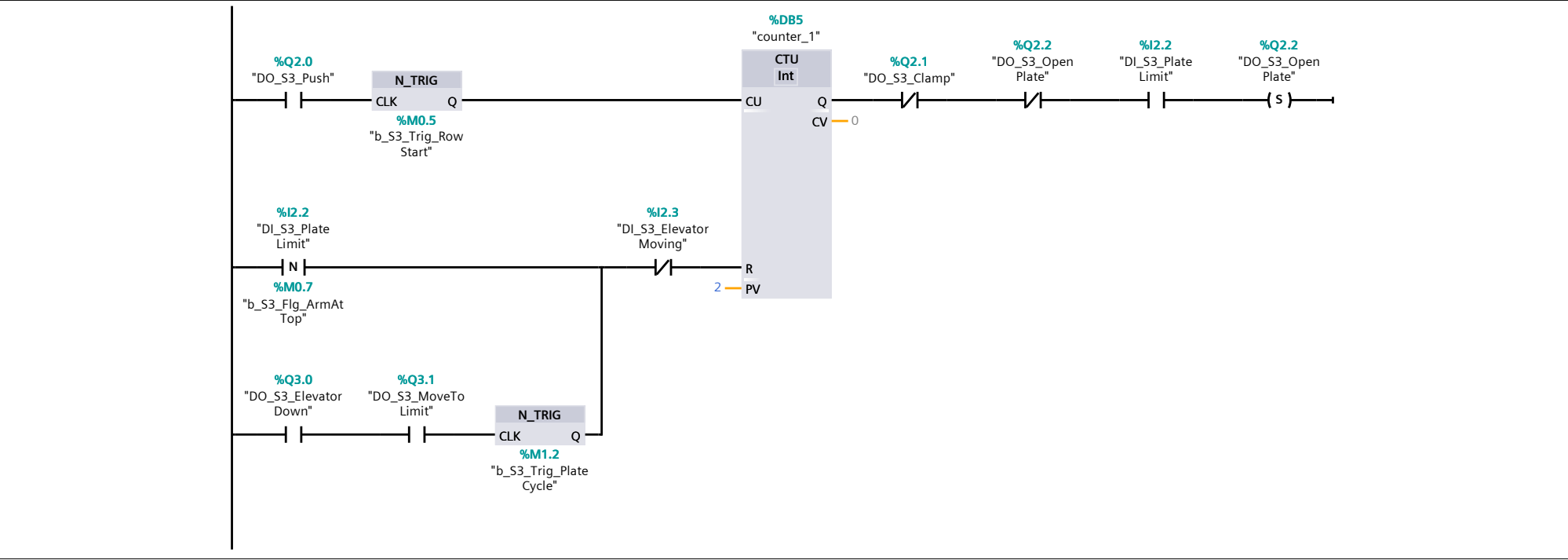
Network 12: Clear the clamp when the pusher limit is reached

Clear the clamp coil (RESET DO_S3_Clamp) when:
- the clamp is energised (DO_S3_Clamp), AND
- the pusher limit sensor is reached (DI_S3_PusherLimit).
The clamp opens once the pusher has finished its travel.



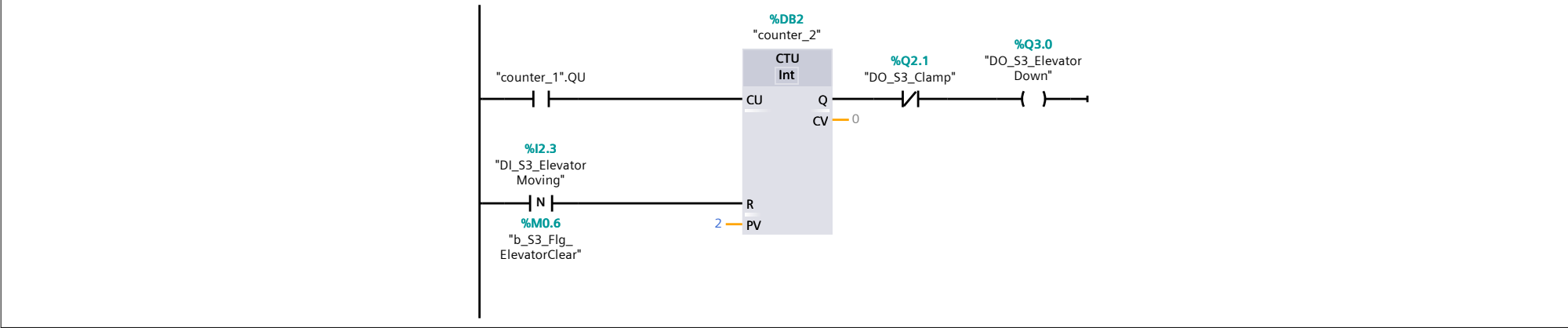
Network 13: Clamp/plate cycle — counter 1 step

Complex step that runs the clamp/plate cycle (counter_1). Triggered by the falling edge of push (N_TRIG b_S3_Trig_PlateCycle). The conditions ensure the cycle only runs when:
- the plate is at its limit (DI_S3_PlateLimit),
- the elevator is not moving (NOT DI_S3_ElevatorMoving),
- the arm is at the top (b_S3_Flg_ArmAtTop), AND
- the elevator is down and the move-to-limit coil is on.
The counter (Cnt_S3_ClampCycle) counts the cycle iterations; when it reaches PV=2 the cycle ends. Then the clamp is reset, the open-plate flag is reset, and the plate is opened (SET DO_S3_OpenPlate) when the plate limit is no longer reached.



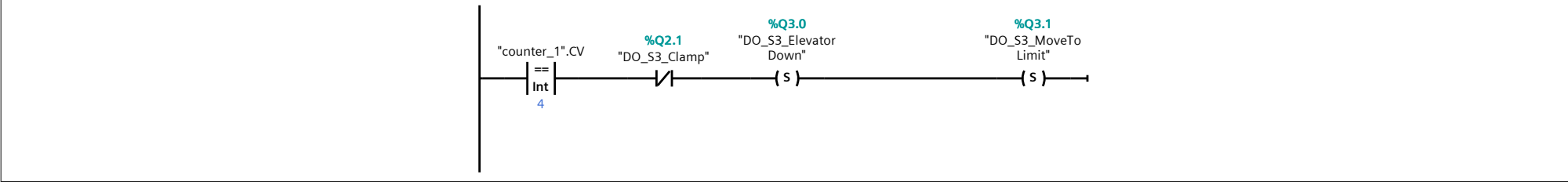
Network 14: Start counter 2 when counter 1 finishes

When counter 1 reaches its end (Cnt_S3_ClampCycle.Q) AND the elevator is not moving (NOT DI_S3_ElevatorMoving) AND the elevator-clear flag is set (b_S3_Flg_ElevatorClear), start counter 2 (Cnt_S3_LowerStep). Also clear the clamp coil (RESET DO_S3_Clamp) AND set the elevator-down coil (SET DO_S3_ElevatorDown) so the elevator begins to lower.



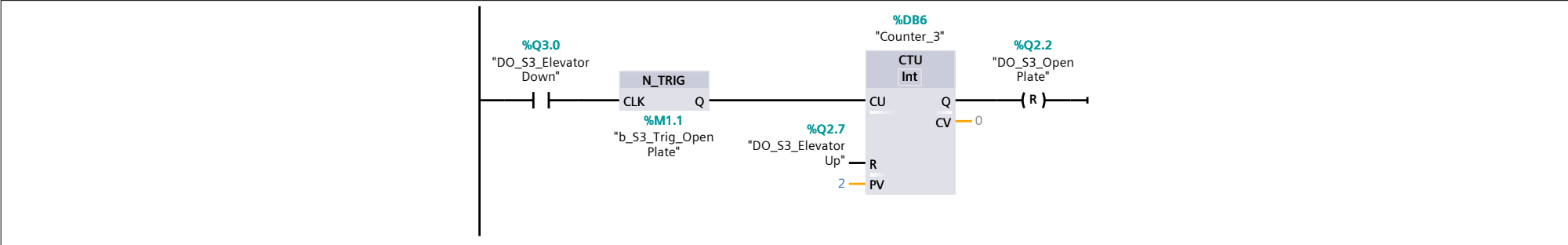
Network 15: Drive the elevator down and to its lower limit

When the clamp-cycle counter (Cnt_S3_ClampCycle) reaches CV=4 AND the clamp is not energised (NOT DO_S3_Clamp), set BOTH the elevator-down coil (SET DO_S3_ElevatorDown) AND the move-to limit coil (SET DO_S3_MoveToLimit). The elevator drives down to its lower limit position.



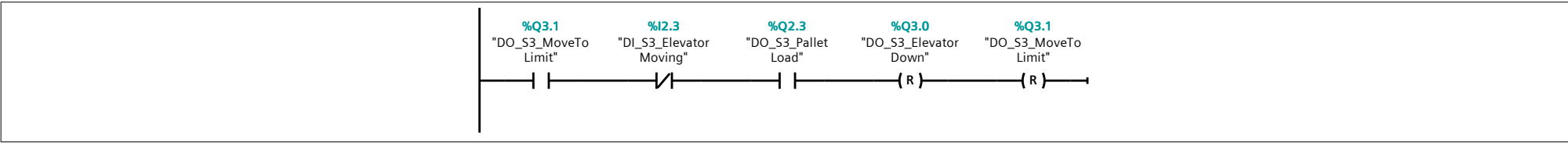
Network 16: Open the plate when the elevator is fully down

When the elevator-down coil is energised (DO_S3_ElevatorDown) AND the falling edge of the open-plate trigger fires (N_TRIG b_S3_Trig_OpenPlate) AND the elevator-up coil is off (NOT DO_S3_ElevatorUp), increment counter 3 (Cnt_S3_PlateOpen) AND reset the open-plate coil (RESET DO_S3_OpenPlate). The plate is opened to release the loaded row onto the pallet.



Network 17: Final reset — when the arm has finished moving

Final cleanup. When the move-to-limit coil is energised (DO_S3_MoveToLimit) AND the elevator is not moving (NOT DI_S3_ElevatorMoving) AND the pallet is loading the outfeeder belt (DO_S3_PalletLoad), reset BOTH the elevator-down coil (RESET DO_S3_ElevatorDown) AND the move-to-limit coil (RESET DO_S3_MoveToLimit). The station is now back to its idle state.



Network 18: Start the outfeed roller when a pallet exits

When a finished pallet reaches the exit sensor (DI_S3_PalletExit) AND the remover coil is energised (SET DO_S3_Remover), start the outfeed roller B (SET DO_S3_Roller4m_B). The pallet is carried away and the remover deletes it from the scene.

